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DEPARTMENT: COMPUTER SCIENCE

LEVEL: 100

COURSE: CSC 104 HARDWARE PRACTICAL

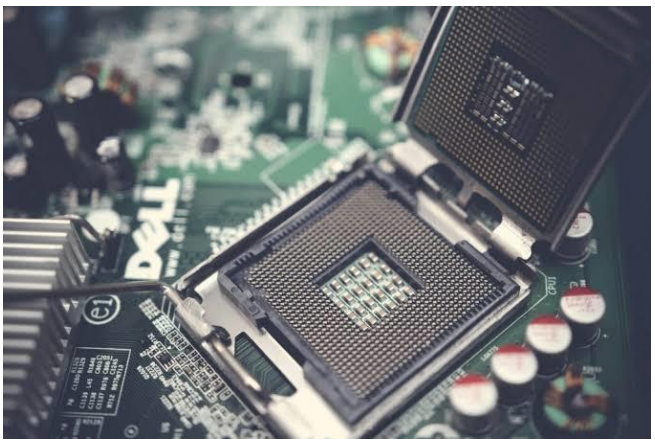
ASSIGNMENT:

1. Write about Microprocessors and RAM
2. Write on different types of processors and their functions

MICROPROCESSORS

Microprocessor is a computer processor that is found in most modern personal computers, smartphones, and other electronic devices. It is a central processing unit (CPU) that performs most of the processing tasks in a computer system. The microprocessor is a key component of a computer, as it controls the fetching, decoding, and execution of instructions that are stored in memory. You can say that microprocessor is used as the brain of the computing devices which control overall execution and operations. The development of microprocessors has played a significant role in the evolution of computers and has made it possible for them to become smaller, faster, and more powerful over time. It works extremely fast, processing billions of instructions per second. Modern microprocessors have multiple cores, allowing them to handle several tasks at once.

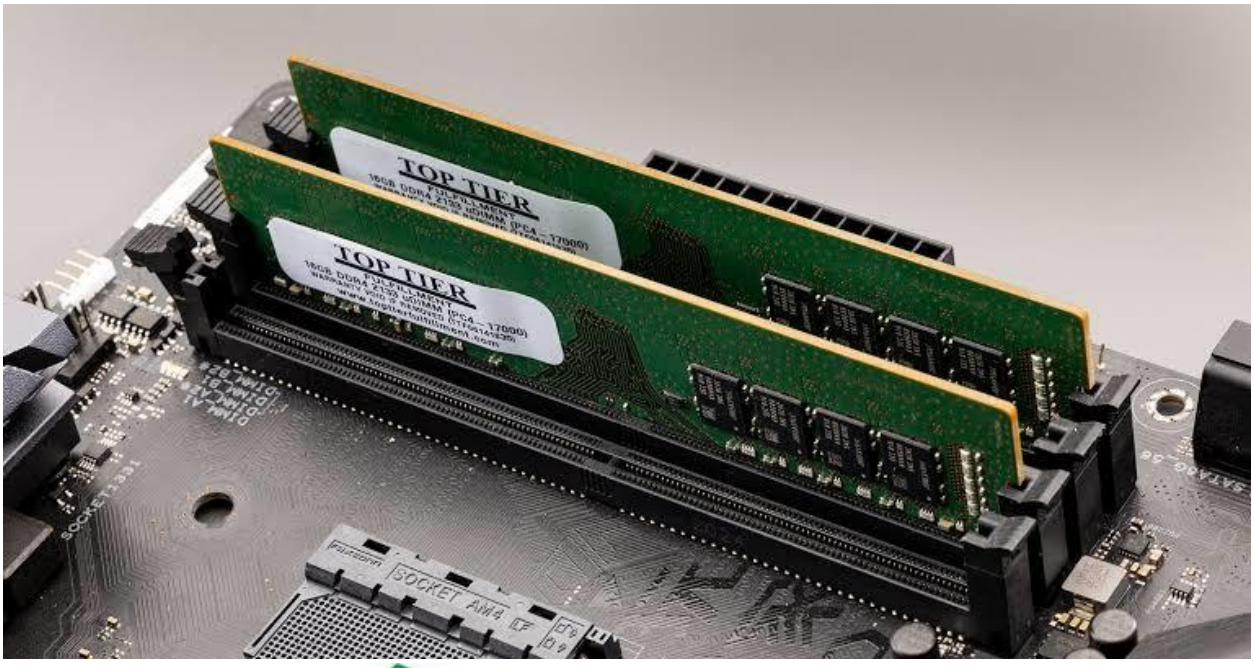




RAM (RANDOM ACCESS MEMORY)

While the microprocessor does the thinking, RAM (Random Access Memory) holds the data it needs. Imagine RAM as a workspace where a student places books while studying. If the table is large (more RAM), the student can keep more books open at once, making learning faster. If the table is small (less RAM), they have to keep putting books away and bringing them back, slowing

things down. RAM is volatile memory, meaning it only stores data temporarily. When the computer is turned off, everything in RAM is erased. That's why files need to be saved to permanent storage (like a hard drive or SSD).





TYPES OF PROCESSORS AND THEIR FUNCTIONS

1. **General-Purpose Processors (CPU)**

These are the most common processors found in computers and laptops. They handle a wide range of tasks, from browsing the internet to running complex software. Examples include: Intel Core series (i3, i5, i7, i9) – Used in personal computers for general tasks.



2. Graphics Processing Unit (GPU)

A GPU is specialized for handling images, videos, and gaming. Unlike a CPU that handles a variety of tasks, a GPU focuses on rendering visuals. It's like an artist who only works on painting rather than doing multiple jobs. Examples include: NVIDIA GeForce & AMD Radeon – Used in gaming and video editing.

Integrated GPUs – Built into the CPU, handling basic graphics tasks.



3. Digital Signal Processor (DSP)

A digital signal processor (DSP) is a specialized microprocessor designed to perform high-speed mathematical operations on digital signals, which are sequences of digital data representing physical quantities such as sound, images, and other sensory information. DSPs are used in a range of applications that require fast processing of digital signals, including audio and video processing, telecommunications, and scientific computing.

Function: Handles real-time signal processing, such as audio, video, and radar signals. **Where Used:** Smartphones (for audio processing), Medical imaging (MRI, CT scans), Voice recognition systems (Siri, Google Assistant), etc



4. **Field-Programmable Gate Array (FPGA)**

Unlike ASICs (which are fixed), FPGAs can be reprogrammed to change how they function, making them useful for testing new technology.

Function: A flexible processor that can be programmed for different applications.
Where Used: Aerospace and defense systems, AI and robotics, 5G and telecommunications



5. **Tensor Processing Unit (TPU)**

TPUs are much faster than CPUs and GPUs when it comes to training AI models.

Function: Designed specifically for machine learning and artificial intelligence.

Where Used: Google AI projects (e.g., Google Photos, Google Translate)**,** Self-driving cars**,** Deep learning models



6. Neuro-processing Unit (NPU)

NPUs learn from patterns, improving performance in tasks like facial recognition and speech processing.

Function: Mimics the human brain's neural network for faster AI computations.

Where Used:

AI-powered smartphones (Huawei Kirin NPU, Apple Neural Engine), Advanced robotics, Smart assistants (Alexa, Google Assistant)



7. **Application-Specific Integrated Circuit (ASIC)**

Unlike general-purpose CPUs, ASICs are optimized for one specific function, making them highly efficient for that task

Function: Designed for a single task, like cryptocurrency mining or video encoding. **Where Used:** Bitcoin mining (ASIC miners), AI and machine learning (Google's TPU), Networking hardware (routers, modems)

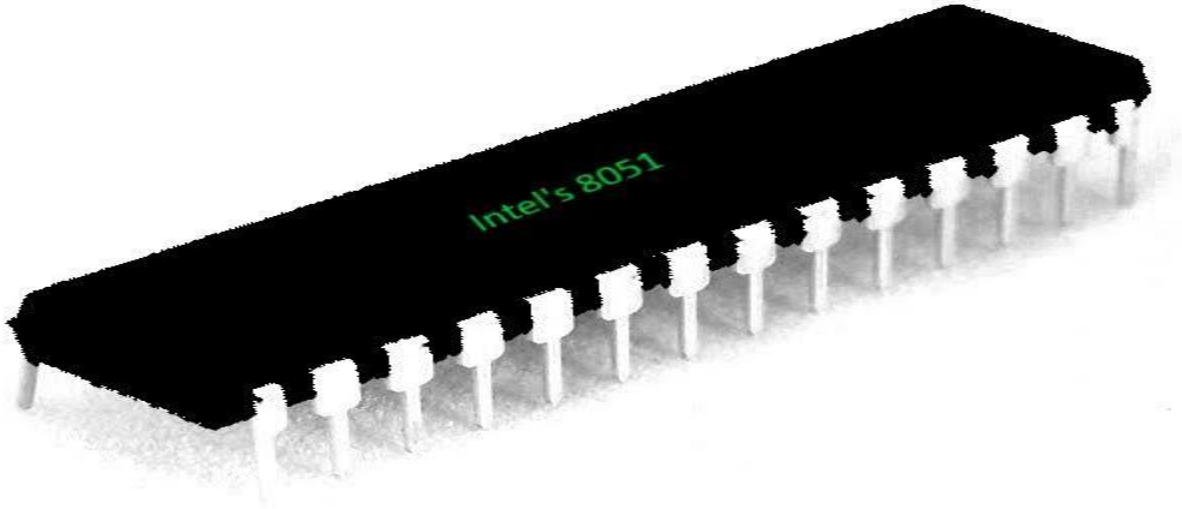


8. Microcontrollers

Microcontrollers are small, low-power processors used in embedded systems like washing machines, traffic lights, and smart home devices. They don't need high power because they perform simple, repetitive tasks. Popular examples:

ARM Cortex series – Found in smartphones and smartwatches.

Arduino & PIC microcontrollers – Used in robotics and automation.



9. Server Processors

These are heavy-duty processors designed for cloud computing and handling large amounts of data. They work like factory managers, ensuring multiple tasks are completed efficiently. Examples include:

Intel Xeon & AMD EPYC – Used in data centers and high-performance computing.


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